Daniel J. Finnegan

ps2fino.com, LinkedIn dr-daniel-j-finnegan-50727620, ORCID 0000-0003-1169-2842

Research interests: Virtual worlds, chattable avatars/NPCs, multisensory perception, head mounted displays, spatial audio interaction, public experiences & installations

https://ps2fino.github.io/documents/Academic_CV.pdf https://ps2fino.github.io/documents/teaching-research-statem pdf

SUMMARY

An academic working in Human Centred AI, Virtual and Mixed Reality (VR & XR), Serious Games, Accessibility, and Spatial Audio Interaction. Work that reaches across disciplinary boundaries, published in rigorously peer reviewed conferences and journals, with grant income commensurate to the career.

EDUCATION

University of Bath, UK September 2017

EngD in Digital Media *Compensating for Distance Compression in Virtual Audiovisual Environments*
Prof Eamonn O'Neill and Prof Michael Proulx
Prof Stephen Payne and Dr Betty Mohler https://ps2fino.github.io/documents/Daniel_J._Finnegan-EngD-Thesis.pdf

University College Dublin, Ireland June 2012

BSc in Computer Science *Object Detection and Tracking in Images and Point Clouds*
Prof Eleni Mangina
3.83/4.0
Best Final Year Project (School of Computer Science); Best Project (Deloitte) https://ps2fino.github.io/documents/Daniel_J._Finnegan-Thesis.pdf

ACADEMIC PRACTICE & REVIEWING

British Computing Society Human-Computer Interaction Conference 2025 2025

General Chair, BCS HCI 2025 Overall responsibility for the conference planning and execution. I led a committee of 21 internationally recognised peers. The conference had a 300% increase in attendance over the previous year. https://www.cardiff.ac.uk/news/view/2988529-cardiff-hosts-bcs-hci-conference-2025

ACM Conference on Human Factors in Computing Systems 2026 2026

Associate Chair, ACM CHI 2026 CHI is my field's top venue; 2026 had 6,730 submissions.

UKRI ESRC 2024 Present

Peer Review College Member

Foundations of Digital Games 2021 2021 2021

Program Committee, Games Beyond Entertainment and Game Education http://www.fdg2021.org/#games-beyond-entertainment-and-game-education

ISMAR 2020 2020 2020

Invited mentorship, Doctoral Consortium https://ismar20.org/call-for-doctoral-consortium/

Proceedings of the ACM on Computer Human Interaction	**Peer Reviewer**
ACM SIGGRAPH VRCAI 2019 2019	**Peer Reviewer** https://vrcai.siggraph.org/committee/
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 2018 2018	**Peer Reviewer**
Experimental Brain Research 2025 2025	**Peer Reviewer**
Interacting with Computers 2025 2025	**Peer Reviewer**
ACM CHI PLAY 2023 2023	**Peer Reviewer**
IFIP TC13 INTERACT 2021 2021 2021	**Peer Reviewer**
Frontiers	**Journal Reviewer**
International Journal of Human-Computer Studies, Elsevier	**Journal Reviewer**
Academic Publishing	**Journal Reviewer**
Public Library of Science	**Journal Reviewer**
Electronics and Telecommunications Research Institute	**Journal Reviewer**
International Journal of Game-Based Learning, IGI Global	**Journal Reviewer**
IEEE Transactions on Parallel and Distributed Systems	**Journal Reviewer**
ACM Transactions on Perception	**Journal Reviewer**
Springer Nature: Scientific Reports	**Journal Reviewer**
Springer Virtual Reality	**Journal Reviewer**
International Journal of Human Computer Interaction	**Journal Reviewer**
International Journal of Gerontology and Social Sciences	**Journal Reviewer**
IEEE Transactions on Human Machine Systems	**Journal Reviewer**

PHD SUPERVISIONS

University of Bath 25/Mar/2026	**Dr Michael Colman** Dr Josie Millar; Dr Bhagyashree Patil; **Dr Daniel J. Finnegan**; Prof Danaë Stanton Fraser; Prof Ailsa Russell Prof Abbie Jordan and Dr Fiona Challacombe Passed with minor corrections *The development and acceptability of a novel virtual reality (VR) tool for the treatment of obsessive compulsive disorder (OCD)*

PHD EXAMINATIONS

Lancaster University School of Computing and Communications, Faculty of Science and Technology 18/Mar/2026	**Dr Abeer Alotaibi** Prof Corina Sas Dr Bryan Williams Prof Tracy Hall
University of Bristol School of Computer Science, Faculty of Science and Engineering 03/Jul/2025	**Dr Richard Grafton** Prof Anne Roudaut Prof Atau Tanaka Prof Seth Bullock https://research-information.bris.ac.uk/ws/portalfiles/portal/474231966/Final_Copy_2025_09_30_Grafton_R_J_PhD.pdf
University of Glasgow School of Education, College of Social Sciences 22/Aug/2023	**Dr Miaomiao Ding** Prof Ellen Boeren Dr Lavinia Hirsu Prof Catherine Lido https://theses.gla.ac.uk/84029/1/2023DingPhD.pdf

TEACHING

CM1301	**Principles, Tools, and Techniques for Secure Software Engineering** Module Leader 250+ students, since 2019. In the end-of-module enhancement survey at Cardiff, the module I lead has received these answers to 'I am satisfied with the overall quality of the module': 2024/25 85% neutral-positive; 2023/24 81% neutral-positive; 2022/23 74% neutral-positive. https://data.cardiff.ac.uk/legacy/grails/module/CM1301.html
CMT206	**Human Centric Computing** 80+ students, since 2019. http://handbooks.data.cardiff.ac.uk/module/CMT206.html
CM2101	**Human Computer Interaction** 250+ students, 2019 - 2024. http://handbooks.data.cardiff.ac.uk/module/CM2101.html
CM6321	**Emerging Technologies** Module Leader http://handbooks.data.cardiff.ac.uk/module/CM6321.html
COMSC X SHARE	**Summer School**
UGT/PGT projects	**Project supervision** I supervise ~ 12 UGT/PGT students every year. Projects vary across all research groups in the school; e.g., machine learning, serious games, web technologies. My students have previously won best dissertation awards from the school.

QUALIFICATIONS & AWARDS

January 2022	**Fellow of the Higher Education Academy (FHEA)** Advance HE
November 2021	**Professional Development Course for External Examiners** Advance HE
May 2022	**Most Innovative Staff Member of the Year (shortlisted)** ESLA, Cardiff's student union awards programme
November 2018	**Recognising Excellence Scheme** Merit Payment for Exceptional Performance; Awarded by the Faculty of Science at the University of Bath

BOARD ROLES

Wavelength June 2026 Present

Board of Trustees member Large national charity; working with young people and communities; I engaged ProMo in knowledge transfer activities, with an emphasis on how they may scale up AI in their operations. https://wavelength.org.uk

ProMo Cymru February 2023 January 2025

Board of Trustees member Large national charity; working with young people and communities; I engaged ProMo in knowledge transfer activities, demonstrating the positive real world impact of my research. https://www.promo.cymru

SOFTWARE

Lancometer My in-class quiz tool. Using my own system gives access to source code, which serves as a resource for teaching programming and web-app concepts. https://ps2fino.github.io/lancometer.html

ViewfindR A software tool for students, educators, and practising professionals in video journalism: a 3D sandbox environment to practise techniques, creating cost-effective opportunities for training and rehearsals. https://viewfindr.cardiff.ac.uk/login

ENGAGEMENT

University of Sri Jayewardenepura in Sri Lanka

Invited keynote speaker, 5th ICIET Conference workshop Invited by the Dean of the Engineering Faculty. https://iciet.sjp.ac.lk/workshops/

INVITED TALKS

2024

Tech4Good Wales

June 2020

Wales Tech Week Public lecture, Multisensory Perception in Virtual Reality
Presented virtually via zoom on human perception in VR, and how technologies can adapt to make a better fit for human-machine systems. https://ps2fino.github.io/wales-tech-week-2020.html

February 2015

Audio Engineering Society Keynote speech, 56th Conference on Audio for Games http://www.aes.org/events/reports/56thConference.pdf#page=5 https://ps2fino.github.io/documents/aes-56-conference.pdf#page=5

October 2018

Bath Spa University Guest lecture, Using Git and GitHub for Project Management

July 2018

Sherborne Science Cafe Invited lecture, Perceptual Based Design for Virtual Reality Environments https://drive.google.com/file/d/1WXt0zayox9fwx4lq-7tmgBG7p-5L4F52/view

May 2018

Cardiff University Brain Research Imaging Centre Invited lecture, Using LaTeX for Open Science https://sciprogramming.wordpress.com/schedule/

May 2018 **Pint of Science** Public lecture, Spatial Perception in Virtual Worlds https://pintofscience.co.uk/event/super-computers-and-ai-who-is-ruling-who

February 2018 **University of Bath** Guest lecture, Games (Beyond Games) https://docs.google.com/presentation/d/1ZPRRx2RlxGZ6RDP5DUFxipOZuhY0wp2lYbgQ1PECn1g/edit

October 2017 **University of Bath** Guest lecture, Auditory Interfaces

November 2016 **Bath Spa University** Guest lecture, Binaural Game Design and Development

February 2016 **Bath College** Careers Day for A-Level Students

EMPLOYMENT

Cardiff University May 2019 Present **Senior Lecturer (Assoc ~ Asst Professor)** I lead the VR sub group in the Human Centred Computing research section; I am the deputy Director of Learning and Teaching in the school; I am the PGR training officer.

Echo Games October 2018 Present **Co-Director** Echo Games CIC develops 'seriously fun' games and interactive experiences for universities, museums, cultural institutions, and not-for-profit organisations. We nurture emerging talent through workshops, game jams and internships, and provide support to cultural institutions that are looking to enhance their digital offer. https://echogames.co.uk

University of Bath October 2016 April 2019 **Research Associate** Postdoc in the CAMERA research centre at the University of Bath; spatial perception in virtual reality.

GRANTS

2026
£1,499,688.00
UKRI Arts Humanities & Research Council (AHRC) — Co-I
Competitive funding awarded for the Transforming Cultures of Language Learning (TransCoLL) project. Lead for Work Package 2 developing Digital Avatars for language learning. Working with several colleagues across UK and Ireland. PI: Fiona Barclay

2024
£1,700.00
EU COST Action Short Term Scientific Mission — PI
Competitive funding awarded to fund a travel grant to Vilnius, Lithuania to visit the Vilnius Toy Museum for a project on digitally augmented historical board games with Indrė Jovaišaitė-Blaževičienė.

2024
£15,000.00
Cardiff University AHSS College — Co-I
Competitive funding awarded to fund a pilot study exploring chattable avatars for museums. Lead for Work Package 2 on human centred development of Digital Avatars. PI: Yipeng Qin

2022
£10,000.00
UKRI Impact Acceleration Account — PI
Competitive funding awarded to fund a pilot study exploring games for cultural heritage.

2024
£30,000.00
UKRI Impact Acceleration Account — Co-I
Competitive funding awarded to further development of ViewfindR (now available for licensing! Get in touch if you would like to know more).

2023
£7,792.00
UKRI Impact Acceleration Account — PI
Competitive funding awarded to further participate in the Being Human Festival. Our exhibition was titled "In Conversation with the AI".

2022
£800.00

Cardiff University AHSS College — Co-I
Competitive funding awarded to fund a workshop on co-production and serious games. PI: Esther Wright.

2022
£100,000.00

Cardiff University Vice Chancellor Strategic Fund — PI
Competitive funding awarded to fund a PhD scholarship on co-production and serious games, with Dr Esther Wright.

2022
£2,000.00

CUSEIP — Co-I
Competitive funding awarded for two projects investigating virtual and mixed reality as a tool for evaluating gamified learning environments, with Dr Geraldine Latchem-Hastings.

2022
£2,000.00

CUROP — PI
Competitive funding awarded for two projects investigating virtual and mixed reality as a tool for evaluating gamified learning environments, with Dr Geraldine Latchem-Hastings.

2022
£15,000.00

GW4 — Co-I
Competitive funding awarded for to investigate gamified experiences for the Bath Assembly Rooms, with Prof Tim Cole and Prof Eamonn O'Neill.

2021
£23,750.00

Innovation Project Translational Fund — Co-I
Competitive funding awarded from the Higher Education Funding Council for Wales (HEFCW) to build ViewfindR prototype version 2.0, a virtual learning environment for teaching students in video journalism to develop their creative skills and plan for filming.

2021
£1,900.00

UKRI GCRF — Co-I
Competitive funding awarded to conduct a workshop with colleagues in IIT Ropar, India, around IoT and VR in healthcare.

2021
£62,177.00

UKRI — PI
Competitive funding awarded from the UKRI's Economics & Social Science Research Council (ESRC) to explore mixed reality in the context of community building interventions for tackling loneliness and reducing feelings of isolation.

2019
£175.00

Heritage Dot Bursary — PI
Competitive funding awarded to present work on developing Serious Games for a Social Change at the Heritage Dot Conference at the University of Lincoln, June 2019

2022
£45,000.00

ESRC Impact Acceleration Account — Co-I
Competitive funding awarded from Cardiff University's IAA scheme for Mark IV and subsequent Mark V prototype development of ViewfindR. ViewfindR is now a commercial project licensed to 5+ institutions across the UK and Europe by UC3, Cardiff University's IP holding company.

2022
£10,000.00

Bristol and Bath Research and Development (Trailblazer Award) — PI
Competitive funding awarded from the Bristol and Bath research and development scheme to create Unlock Bath: A digital Escape Room exploring the interconnected history of the city of Bath in cooperation with several local museums.

Total grant capture: £1,826,982.00

PUBLICATIONS

2026 **"Am I now masking, even with a bot?": Pseudo-Social Bonding between Autistic Adults and LLMs** *BritCHI 2026*
Cogley, E., Matthews, T., Finnegan, D. J.https://dx.doi.org/10.14236/ewic/BRITCHI26.46

2026 **Comparative effects of cooperative and competitive social models on motivation and physiological engagement in VR exergaming** *Frontiers in Virtual Reality*
Gao, G., Lynch, G., Finnegan, D. J., Verdezoto, N. D., Barathi, S.https://www.frontiersin.org/journals/virtual-reality/articles/10.3389/frvir.2026.1897543/full

2026 **Novel virtual reality methods for the treatment of OCD: a user-centred design approach** *JMIR XR & Spatial Computing*
Colman, M., Millar, J. F. A., Patil., B., Finnegan, D. J., Kristoffersen, M., Fraser, D. S.https://dx.doi.org/10.2196/92410

2026 **Deaf Play: Social Play and Signed Communication among Deaf Gamers in Co-Located Environments** *BritCHI 2026*
Curtis, M., Gould, J. J. S., Finnegan, D. J.https://orca.cardiff.ac.uk/id/eprint/188609/1/494-Curtis-BCSHCI26.pdf

2026 **Empowering Touchscreen Accessibility for Individuals with Multiple Sclerosis and Dexterity Impairments: A User-Centered Approach to Touch-Free Gestures** *Universal Access in the Information Society*
Bamoallem, R., Loizides, F., Finnegan, D. J., , Al-Haqbani, A., Al Dagher, S.https://ps2fino.github.io/uuia-2026

2026 **Supervisory Control via AR for Teleoperation under Communication Delays: An In-Space Assembly Use Case** *International Conference on Social Robotics*
Stavropolous, I-M., Kucukyilmaz, A., Martinez, C., Finnegan, D. J., Hernandez, J. H.https://orca.cardiff.ac.uk/id/eprint/187027/

2025 **An Integrative Process for the Design of Seriously Fun Games** *Games and Learning Alliance*
De Angeli, D., Finnegan, D. J., Scott, L.https://link.springer.com/chapter/10.1007/978-3-032-11043-5_53

2025 **'Chattable' Avatars: Using LLMs to Power Visitor Engagement with Historical Persons** *BCS HCI 2025*
Jiang, Z., Qin, Y., Finnegan, D. J.https://www.scienceopen.com/hosted-document?doi=10.14236/ewic/BCSHCI2025.10

2025 **HF-MAI: Touchless Mid-Air Interaction for Individuals with Dexterity Impairment** *IFIP TC13 INTERACT 2025*
Bamoallem, R., Loizides, F., Finnegan, D. J., AlSunaidi, H.https://orca.cardiff.ac.uk/id/eprint/180300/

2025 **Playful Heritage: Video Game Creation as Interdisciplinary Historical Engagement** *Journal of Cinema and Media Studies*
Wright, E., and Finnegan, D. J.https://quod.lib.umich.edu/j/jcms/18261332.0064.909/--playful-heritage-video-game-creation-as-interdisciplinary?rgn=main;view=fulltext

2024 **Advantages of Friend-Modelled Social Interactive Feedforward for VR Exergaming** *ACM CHIPLAY 2024* (Expected Submission to REF)
Barathi, S. C., Finnegan, D. J., Proulx, M. J., O'Neill, E., Lutteroth, C.https://dx.doi.org/10.1145/3677103

2024 **A systematic review and narrative synthesis of the use and effectiveness of extended reality technology in the assessment, treatment and study of obsessive compulsive disorder** *Journal of Obsessive-Compulsive and Related Disorders*
Colman, M., Millar, J., Patil, B., Finnegan, D. J., Russell, A., Higson-Sweeney, N., Aguiar, M., Stanton Fraser., D.https://orca.cardiff.ac.uk/id/eprint/170514/

2024 **Potential factors contributing to observed sex differences in Virtual Reality induced sickness** *Experimental Brain Research* (Expected Submission to REF)
Bannigan, G. M., de Sousa, A., Scheller, M., Finnegan, D. J., Proulx, M. J.https://orca.cardiff.ac.uk/id/eprint/164470/8/s00221-023-06760-0.pdf

2023 **Tackling Loneliness and Isolation in older adults with Virtual Reality: How do we move forward?** *Gerontology and Geriatric Medicine*
Finnegan, D. J., Campbell, S.https://journals.sagepub.com/doi/10.1177/23337214231186204

2022 **Understanding spatial, semantic and temporal influences on audiovisual distance compression in virtual environments** *PsyArXiv*
https://dx.doi.org/10.31234/osf.io/u3wc2

2022 **Unsettling Play: Perceptions of Agonistic Games** *ACM Journal of Computing and Cultural Heritage Special Issue on Culture Games* (Expected Submission to REF)
De Angeli, D., Finnegan, D.J., Scott, L., O'Neill, E.https://orca.cardiff.ac.uk/136295/

2021 **Does Mixed Reality have a Cassandra Complex?** *Frontiers in Virtual Reality*
Finnegan, D. J., Zoumpoulaki, A., Eslambolchilar, P.https://dx.doi.org/10.3389/frvir.2021.673547

2021 **'Did you see that!?' Enhancing the Experience of Sports Media Broadcast for Blind People** *IFIP TC13 INTERACT 2021*
Goncu, C., & Finnegan, D. J.https://dx.doi.org/10.1007/978-3-030-85623-6_24

2021 **A Virtual Reality Application of the Rubber Hand Illusion Induced by Ultrasonic Mid-Air Haptic Stimulation** *ACM Transactions on Applied Perception*
Salagean, A., Hadnett-Hunter, J. E., Finnegan, D. J., de Sousa, A., Proulx, M.https://orca.cardiff.ac.uk/id/eprint/143639/

2020 **Immersive virtual environments and embodied agents for e-learning applications** *PeerJ Computer Science*
Fitton, I. S., Finnegan, D. J., Proulx, M. J.https://orca.cardiff.ac.uk/135879/

2020 **Sacred Springs: Teaching Children Local History via a Game Jam** *World's Most Inclusive Participatory Design Workshop @ ACM IDC 2020*
De Angeli, D., Finnegan, D. J., Scott, L.documents/game-jam-inclusive-pd-2020.pdf

2018 **Agonistic Games: Multiperspective and Unsettling Games for a Social Change** *CHI PLAY '18 Extended Abstracts*
De Angeli, D., Finnegan, D. J., Scott, L., Bull, A., O'Neill, E.documents/chiplay-2018.pdf

2018 **HandMap: Robust Hand Pose Estimation via Intermediate Dense Guidance Map Supervision** *ECCV 2018: Proceedings of the 15th European Conference on Computer Vision* (Submitted to REF)
Wu, X., Finnegan, D. J., O'Neill, E., Yang, Y.documents/eccv-2018.pdf

2018 **Interactive Feedforward for Improving Performance and Maintaining Intrinsic Motivation in VR Exergaming** *CHI 18': Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Submitted to REF)
Barathi, S. C., Finnegan, D. J., Farrow, M., Whaley, A., Heath, P., Buckley, J., Dowrick, P. W., Wünsche, B. C., Bilzon, J. L. J., O'Neill, E. & Lutteroth, C.documents/chi-2018.pdf

2017 **An Approach to Reducing Distance Compression in Audiovisual Virtual Environments** *2017 IEEE 3rd VR Workshop on Sonic Interactions for Virtual Environments (SIVE)*
Finnegan, D. J., O'Neill, E., & Proulx, M. J.documents/sive-2017.pdf

2016 **Compensating for Distance Compression in Audiovisual Virtual Environments Using Incongruence** *CHI '16: Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* documents/chi-2016.pdf

2014 **Reindeer & Wolves: Exploring Sensory Deprivation in Multiplayer Digital Bodily Play** *CHI PLAY '14: Proceedings of the First ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play* documents/chiplay-2014.pdf